Run the project on Linux

Ubuntu, Debian, Pop!_OS, Fedora · Sbusiso Shinga · 20 October 2026

At the end you have a pipeline that collects job postings from public APIs, an SFTP drop and a PostgreSQL CRM, streams them through Kafka into a data lake, transforms them with Spark, models them with dbt, and answers "what are employers asking for?" from a REST API and a dashboard.

Repository: https://github.com/Sbusiso-tech/job-market-intelligence-platform

1. Requirements

Tool	Minimum	Check
Docker Engine	24 or newer	`docker --version`
Docker Compose plugin	v2.24 or newer	`docker compose version`
make	any recent	`make --version`
Free RAM	about 6 GB	Spark needs the headroom
Free disk	about 8 GB	images plus the lake

Your user must be in the `docker` group (`sudo usermod -aG docker $USER`, then log out and back in).

2. First run

```
git clone https://github.com/Sbusiso-tech/job-market-intelligence-platform.git
cd job-market-intelligence-platform
cp .env.example .env
make up
make pipeline
make top-skills
```

`make up` builds the app image and starts Kafka, RustFS, both PostgreSQL databases, Redis, the SFTP server, the lake sink, the realtime consumer and the API. The first build downloads Spark and can take several minutes. `make pipeline` then ingests, drains Kafka into the lake, runs Spark and builds the dbt marts.

Himalayas, Jobicy, Remotive and Arbeitnow need no keys. Adzuna and JSearch are skipped until you put keys in `.env`. The partner CSV and the CRM are simulated: `make partner-drop` and `make crm-activity`.

3. Where to look

What	Where	Login
API docs	http://localhost:8000/docs	
Kafka UI	http://localhost:8081	
RustFS console	http://localhost:9001	rustfsadmin / rustfsadmin
Airflow	http://localhost:8080 after `make airflow`	admin / admin

Metabase	http://localhost:3000 after `make metabase`	you set this on first visit
Warehouse	localhost:5432, database jobmarket	jobmarket / jobmarket
Recruitment CRM	localhost:5433, database crm	crm / crm
Partner SFTP	sftp://localhost:2222	partner / partner

For Metabase, connect to host `postgres`, port 5432, database `jobmarket`, schema `marts`.

4. Optional pieces

```
make airflow       # DAG job_market_pipeline, 06:00 and 18:00
make cdc           # Debezium on crm.vacancies, then GET /jobs/latest
make partner-drop  # today's partner CSV
make crm-activity  # recruiters open, reprice, fill and close vacancies
make down          # stop, keep data
make clean         # stop and delete volumes
```

`make cdc` waits until Kafka Connect is up, then registers the connector. The CRM database must be the one started by this compose file, because it runs with `wal_level=logical`.

5. Checks

```
curl -s localhost:8000/health
curl -s "localhost:8000/skills/top?limit=5" | python3 -m json.tool
make test
```

GitHub Actions runs the same lint, unit tests and `dbt parse` on every push to `main`.